

KUO, CHEN CHIH 郭晨志

M.S. in Chemical Engineering • Licensed environmental specialist

Seeking EHS Engineer

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Xinzhuang, New Taipei • Open to relocation / shifts

Profile

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (**wastewater, waste, indoor air**), OSH exams scheduled; hands-on with **hazardous materials** (lithium metal, organic-solvent electrolytes, faulty Li-SOCl₂ cells), lab safety and **thermal hazard analysis** (TGA / DSC).

Experience

- 2025 – Present

Sustainable Electrochemical Energy Development (SEED) Center, NTUST • MOE Featured Areas Research Center
Lab Equipment Procurement & Asset Maintenance Manager
 - Procurement (>NTD 2M):** Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes, low-temperature baths and TGA platinum pans; also replace instrument consumables.
 - Lab safety:** monitor glovebox O₂ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak.
 - Assembled cells and prepared electrodes in a battery dry room, working in cleanroom garments under clean-area gowning and entry procedures.
 - Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl₂ cells.
 - TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's.
 - TGA workflow:** changed how samples are prepared, raising sample mass and precision, reducing platinum-pan contamination and extending pan life; adopted the vendor's liquid-sample method and set PPE, ventilation and booking rules.
 - Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.
- Summer 2022

New Taipei City Dept. of Urban Development
Summer Intern, Housing Development Section
 - Reviewed rental subsidy eligibility, cross-checking property, income, and insurance records across agencies (tax bureau, land registry, NHI); handled frontline public inquiries and disputes.
- 2021 – 2024

Tunghai University Student Housing / GEPT
Resident Assistant (2 yrs) / Exam Staff (part-time)
 - Oversaw safety and order in a 100-resident dormitory, mediating disputes and tracking facility safety issues with the dorm supervisor; as part-time exam staff, handled test-paper check-in with zero errors.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

Screening methodology

Designed full-stage test protocols, experimental workflows and statistical validation models.

Database & SOP

Established the test database and standardized high-throughput screening procedures (SOP).

Advanced analysis

EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

Training & tech transfer

Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Materials Journal submission — battery thermal stability study Status **Under review**

Co-author Responsible for TGA thermal measurement

Conferences & Training

Presenter 2026 TwiChE Annual Meeting

2026/12/05–06 • Feng Chia University

Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li // SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability*

Vendor seminar TA Instruments ARES-G3 launch & applications forum

2026/07/17 • National Taiwan University

Vendor lecture TA Instruments TGA lecture

2026/06/03 • National Taiwan University

EHS Licenses

- ✓

Dedicated Wastewater and Sewage Treatment Specialist (Class B)
2026 / 06 • certified • No. GB040231
- ✓

Indoor Air Quality Maintenance Management Specialist
2026 / 08 • certified • No. IA020236
- ✓

Waste Disposal Technician (Class A)
2026 / 10 • certified • No. HA160629
- Occupational Safety and Health Management Personnel (Level B)
2026 / 11 • mid-Nov exam scheduled
- Occupational Safety Management Specialist (Level A)
2027 / 03 • exam scheduled

Education

2025 – 2027 (expected)

National Taiwan University of Science and Technology

M.S. in Chemical Engineering

Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.

2020 – 2024

Tunghai University

B.S. in Chemical & Materials Engineering

Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

Skills & Instruments

EHS fundamentals

- Chemical labelling rules (GHS)
- Explosive limits (LEL / UEL)
- Occupational exposure limits
- Ventilation

Electrochemistry & batteries

- Impedance spectroscopy (EIS)
- Battery aging analysis
- Reliability analysis
- Low-temp electrolytes

Process & operations

- Lithography (nanoimprint / self-assembly)
- Equipment procurement & maintenance

Software

- Aspen Plus V14
- Aspen Dynamics
- Microsoft Office
- OriginPro
- TA Universal Analysis
- TA TRIOS

Instruments

Maintenance

- TGA Q500
- DSC 2500

Operation

- Anton Paar DMA 5002/Lovis 2001
- ARES-G3
- Glovebox
- Synchrotron (TLS 20A / TPS 27A / TPS 45A)

Procurement

- Glovebox O₂ sensor
- DSC sample press
- Hydraulic electrode punch
- Vacuum pump
- Eppendorf Pipette
- Low-temperature bath